ALI AHAD

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ABOUT

PhD in Electrical and Computer Engineering with a focus on systems and software security, specializing in program analysis, reverse engineering, and forensic techniques. My research focuses on utilizing program analysis and LLMs to identify and mitigate vulnerabilities in large-scale systems, and enhance reverse engineering tools for effective forensics.

EDUCATION

University of Maryland – College Park, MD — Defended April 2026

- PhD in Electrical and Computer Engineering

University of Virginia – Charlottesville, VA — August 2023

- MS in Computer Science — GPA: 4.0/4.0

Lahore University of Management Sciences – Lahore, Pakistan — June 2020

- BS in Computer Science — GPA: 3.9/4.0

EXPERIENCE

University of Maryland – College Park, MD — January 2024 – Present

Research Assistant

- Implement and evaluate novel malware analysis and information flow tracking methods; advance reverse engineering technologies to detect hidden malicious code and information leaks in real-world systems.
- Developed FreePart, a partitioning technique for secure data processing (ASPLOS '24), and DeARM, an LLM-driven framework for improving ARM64 decompilation that outperforms SOTA decompilers by 10.2%.

NVIDIA – Santa Clara, CA — May 2024 – August 2024

System Software Intern, Data Center System Security

- Enhanced TPM security for data center systems by implementing and validating DMTF/TCG TPM-over-SPDM protocols, prototyped on Virtual TPM (vTPM) for secure measurements and communications.
- Authored production roadmap and presented prototype to cross-functional leads, guiding transition from virtual proof-of-concept to hardware-integrated security features.

NVIDIA – Santa Clara, CA — September 2023 – January 2024

Software Intern, Security

- Engineered a TPM-based attestation solution for server platform security, implementing EK/AK provisioning, RIM generation, and evidence reporting using the RATS architecture.
- Built a TPM library with tpm2-pytss to streamline attestation workflows and improve security on NVIDIA systems.

University of Virginia – Charlottesville, VA — August 2020 – August 2023

Research Assistant

- Published four papers in top-tier venues in Security/Systems (CCS ’21, FSE ’21, S&P ’22, S&P ’23).
- Led a cross-institutional project resulting in a first-author S&P ’23 publication; mentored two undergraduate interns (Amazon Summer ’22 & Appian Summer ’23), guiding research and project development.

PUBLICATIONS

- **"FreePart: Hardening Data Processing Software via Framework-based Partitioning and Isolation"**: **Ali Ahad**, Gang Wang, Chung Hwan Kim, Suman Jana, Zhiqiang Lin, and Yonghwi Kwon, *In Proc. of the 29th International Conference on Architectural Support for Programming Languages and Operating Systems* (ASPLOS '24).
- **"PyFET: Forensically Equivalent Transformation for Python Binary Decompilation"**: **Ali Ahad**, Chijung Jung, Ammar Askar, Doowon Kim, Taesoo Kim, and Yonghwi Kwon, *In Proc. of the 44th IEEE Symposium on Security and Privacy* (S&P '23).
- **"SwarmFlawFinder: Discovering and Exploiting Logic Flaws of Swarm Algorithms"**: Chijung Jung, **Ali Ahad**, Yuseok Jeon, and Yonghwi Kwon, *In Proc. of the 43rd IEEE Symposium on Security and Privacy* (S&P '22).
- **"Forensic Analysis of Configuration-based Attacks"**: Muhammad Adil Inam, Wajih Ul Hassan, **Ali Ahad**, Adam Bates, Rashid Tahir, Tianyin Xu, and Fareed Zaffar, *In Proc. of the 29th Network and Distributed System Security Symposium* (NDSS '22).
- **"Swarmbug: Debugging Configuration Bugs in Swarm Robotics"**: Chijung Jung, **Ali Ahad**, Jinho Jung, Sebastian Elbaum, and Yonghwi Kwon, *In Proc. of 29th ACM SIGSOFT International Symposium on the Foundations of Software Engineering* (FSE '21).
- **"Spinner: Automated Dynamic Command Subsystem Perturbation"**: Meng Wang, Chijung Jung, **Ali Ahad**, and Yonghwi Kwon, *In Proc. of 28th ACM Conference on Computer and Communications Security* (CCS '21).

RELEVANT COURSES

- **Program Analysis**: Software Analysis, Program Analysis, Compilers
- **Security**: Mobile & IoT Security, Network Security & Privacy, Cyber Forensics
- **Systems**: Computer Architecture, Operating Systems, Digital CMOS VLSI Design
- **Machine Learning**: Intro. to Artificial Intelligence, Machine Learning, Information Theory
- **Networks**: Internet Infrastructure, Network-Centric Computing

TECHNOLOGY

- **Languages**: Python, C, C++, Rust, BASH, Golang, Javascript, Dart
- **Frameworks**: PyTorch, TensorFlow, HuggingFace Transformers, PEFT/LoRA, LLVM, Pandas, React, Flask
- **Reverse Engineering**: Ghidra, IDA, Angr, Uncompyle6, Decompyle3
- **Software Testing**: American Fuzzy Lop (AFL), KLEE, LibFuzzer
- **Infrastructure**: Git, Linux, Docker, SLURM, Wireshark

SKILLS

- **Security & Systems**: Systems & Software Security, Reverse Engineering, Program & Information Flow Analysis, Cyber Forensics, Secure System Design, Threat Modeling, Malware Analysis, Vulnerability Assessment
- **AI/ML & Program Analysis:** LLM Fine-Tuning & Specialization, Binary Analysis, Static & Dynamic Analysis, Algorithm Design & Optimization, Control Flow Analysis, Model Evaluation & Benchmarking
- **Systems Engineering:** Large-Scale Tool Development, Automated Testing & Fuzzing, Software Debugging
- **Research & Communication:** Cross-Functional Collaboration, Technical Roadmapping, Scientific Writing